Recap

Exercise 1:

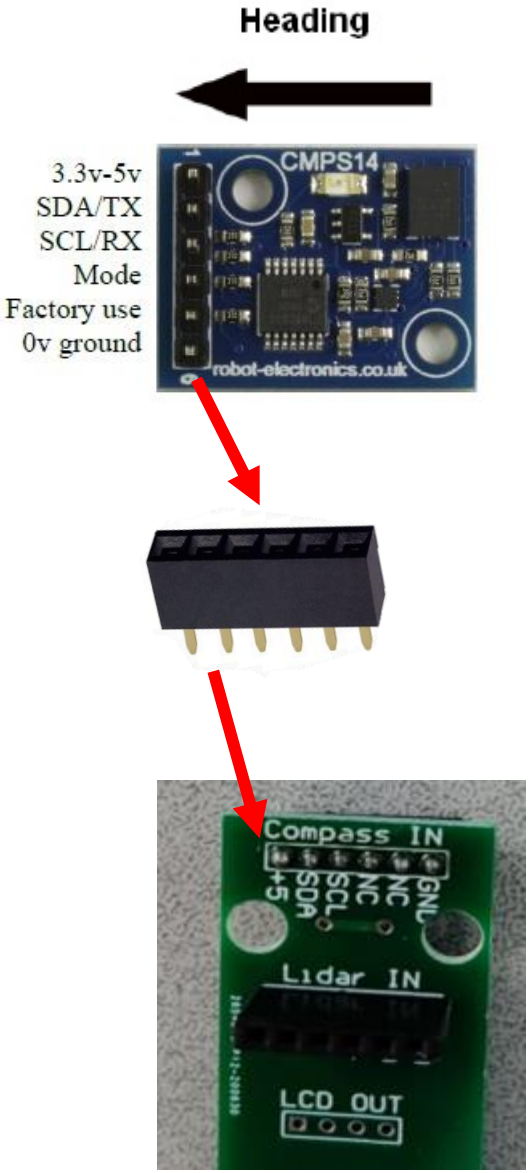
- Create a program that drives the car, adjusting its speed at specific distances according to the following guidelines:
  - drive at 50% speed for the first 10 cm
  - at 25% speed for the next 5 cm,
  - and 75% speed for the final 15 cm.
- The car should stop automatically after reaching the total target distance (30cm).
- Use the pulses from your encoder to determine the distance driven
- Pressing the joystick push button will execute the driving again (once)
  - you can connect the joystick push button to any of the still available ISR supported pins if it isn't connected yet/anymore
    - don't change the encoder pulse pins unless really, really necessary

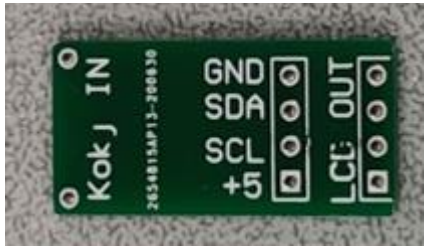


Compass & I²C bus

Exercise 2:

- Connect the compass to your cable set
  - Don’t solder the compass, instead use headers in the Compass IN pin →
  - The heading should point to same direction as your car’s front
  - Place the compass flat on top of your car, not upright!
- Connect the other end to your breadboard and from there to Arduino ↓↓
  - You might need to google what Mega pins are used for I²C (SDA and SCL)

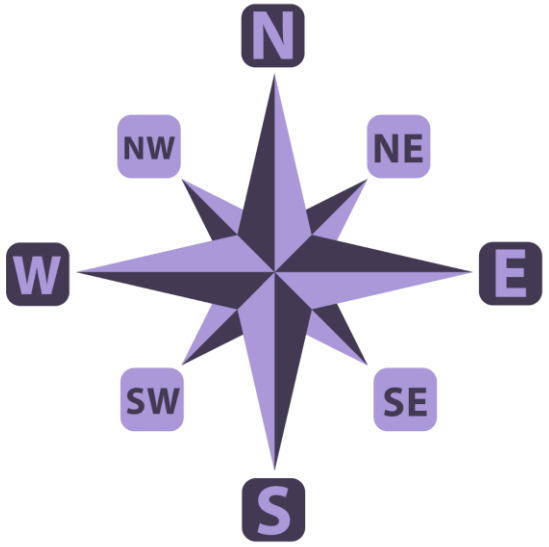




Compass & I²C bus

Exercise 3:

- Create a simple code where you read a raw 8-bit bearing value from the compass.
  - Print the value to Serial monitor
  - Code help in Moodle lecture slides: I2C bus
    - If necessary, download the Wire.h library to your Arduino Ide
      - Tools -> Manage libraries



Compass & Degrees

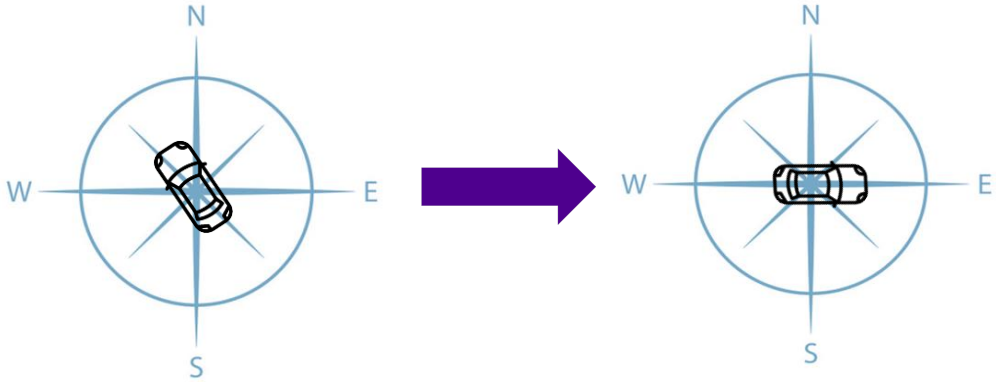
Exercise 4:

- Continue with your code and print the compass bearing value (0-360°) onto your LCD
- Test a few times in different parts of the lab where North(N) is according to your sensor
- Can we trust the readings? Does lab equipment interfere?

- Add directional names (N, NW, W, etc.) next to the degree value on you LCD according to the table values →

Direction	Degree
NE	23 – <68
E	68 – <113
SE	113 – <158
S	158 – <203
SW	203 – <248
W	248 – <293
NW	293 – <338
N	338 – <23

Compass & Turning



Random starting degree

Facing given degree

Exercise 5:

- Create a program that turns your car to face any specified heading (degree) between 0° and 360°.
- Similarly to an earlier driving distance exercise, you can hard code the wanted heading value in the beginning of your code
- The car should rotate until it aligns with the specified heading. If the car is manually turned after reaching the target, it should automatically correct its position to face the correct heading again.
  - Use a small +/- offset to ensure the car doesn't keep rotating
    - for example, if the desired degree is 90 (East), values 89 and 91 are also fine
- Try to keep your Main loop as empty as possible and implement the turning as a new function e.g. void find_heading(int given_degree)

Compass & Encoder

Exercise 6:

- Develop a program that controls your car according to the route guidelines → →
  - Car starts and finishes at the same spot
  - Use Encoder pulses and compass
  - Car's starting heading/direction is random, so your code needs to work which ever direction it starts at
  - Like in exercise 1, pressing the joystick push button will execute the driving again (once)

